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Protein Pair-wise Sequence Alignment Webtool

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1. INTRODUCTION

This webpage provides a web-based tool for pair-wise protein sequence alignment service. Both a GLOBAL alignment using the Needleman-Wunsch algorithm and a LOCAL alignment using the Smith-Waterman algorithm are available. Unlike other ordinary alignments, users may select up to three substitution scoring matrixes with various weight for each matrix.

2. FEATURES

A. Alignment Algorithm

Two algorithms may be used. They employ substitution scoring matrixes instead of simple 'match/mismatch'.

① Needleman-Wunsch: A global alignment algorithm

② Smith-Waterman: A local alignment algorithm

B. Sequences

① Protein sequences may be pasted into the text box, or user can upload a text file containing a protein. Uploading a file has a higher priority than text area

② Non-amino acid characters will cause an error message. However, numeric characters and blank spaces are allowed and will be automatically removed.

③ A FASTA file or a sequence with a preamble line is also accepted.

C. Substitution Scoring Matrixes

① Users may select up to three different matrixes with respective weight.

② A new scoring matrix will be created and used depending on users' selection of matrix and weight

③ Currently standard BLOSUM and PAM matrixes are provided. Additional matrixes are hydrophobicity matrix and size matrix.

④ A weight value can be between 0 and 10. But at least one weight should be greater than zero. These weights are relative values, which mean that 1:1:1 is equal to 10:10:10.

D. Gap penalty

① Gap Open Penalty: To specify the opening gap penalty.
Default value is -10

② Gap Extension Penalty: To specify the gap extension penalty.
Default value is -1

E. Number of Maximum Alignments

① If more than one optimal alignment is found, user may set a limit to display to the higher ranked alignments. The default value is 3.

F. Result via Email

① Users may get the alignment result via email. Regardless of this option, users can see the results on the next display.

G. SAMPLE_SEQ Button

① This button will put two sample sequences into the sequence text boxes. They are variants of Drosophila Shaggy (Glycogen Synthase Kinase 3) proteins.

H. Result Download

① At the bottom of the result page, there will be three links for users to download the result files in text format. Please use right mouse button and save the target.

3. Clickable Buttons

A. **ALIGN** : Begin to align two protein sequences

B. **CLEAR** : Erase the user's input

C. **SAMPLE_SEQ** : Put two sample protein sequences into the text area

4. Links

A. Main Page

① README : This read-me file

② Sample_Result : Sample alignment results with two shaggy proteins

③ L519 Bioinformatics : Simple introduction page to the class

④ Email : Send an email to the author

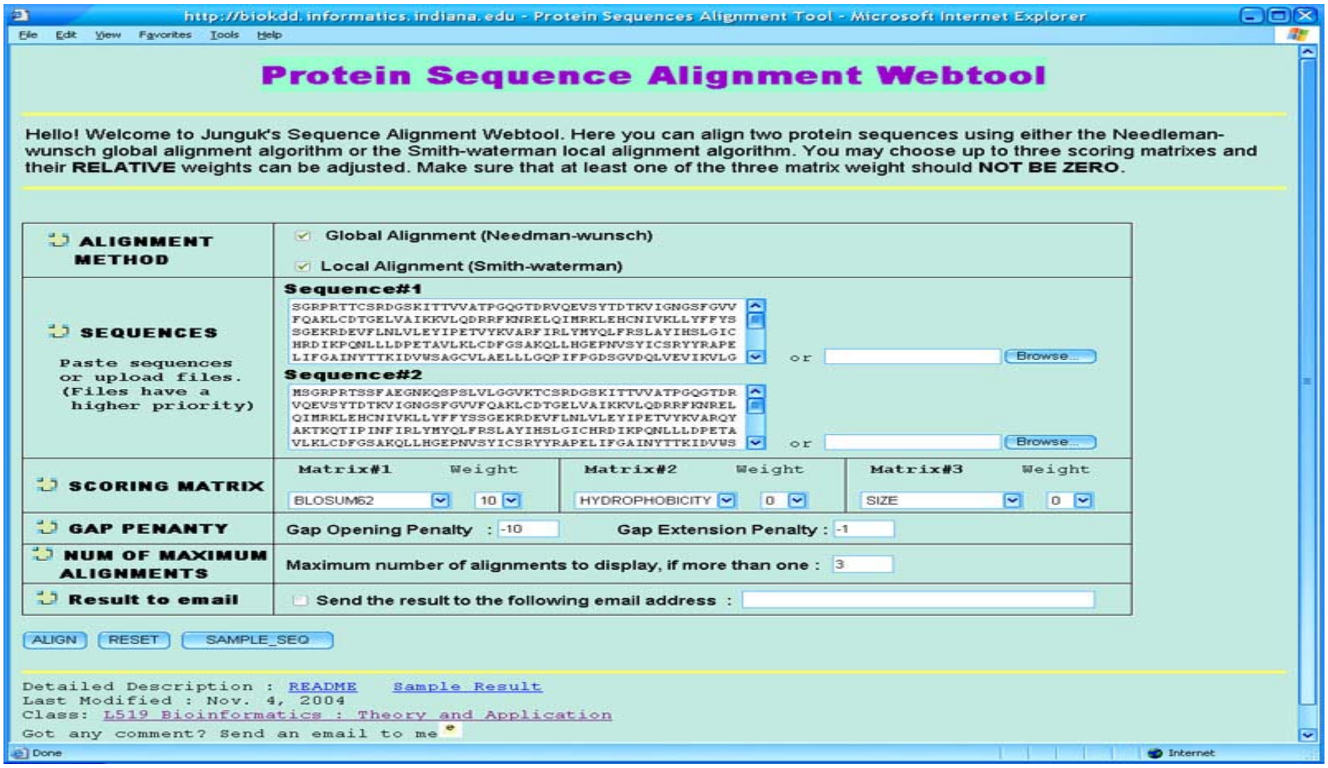
B. Result Page

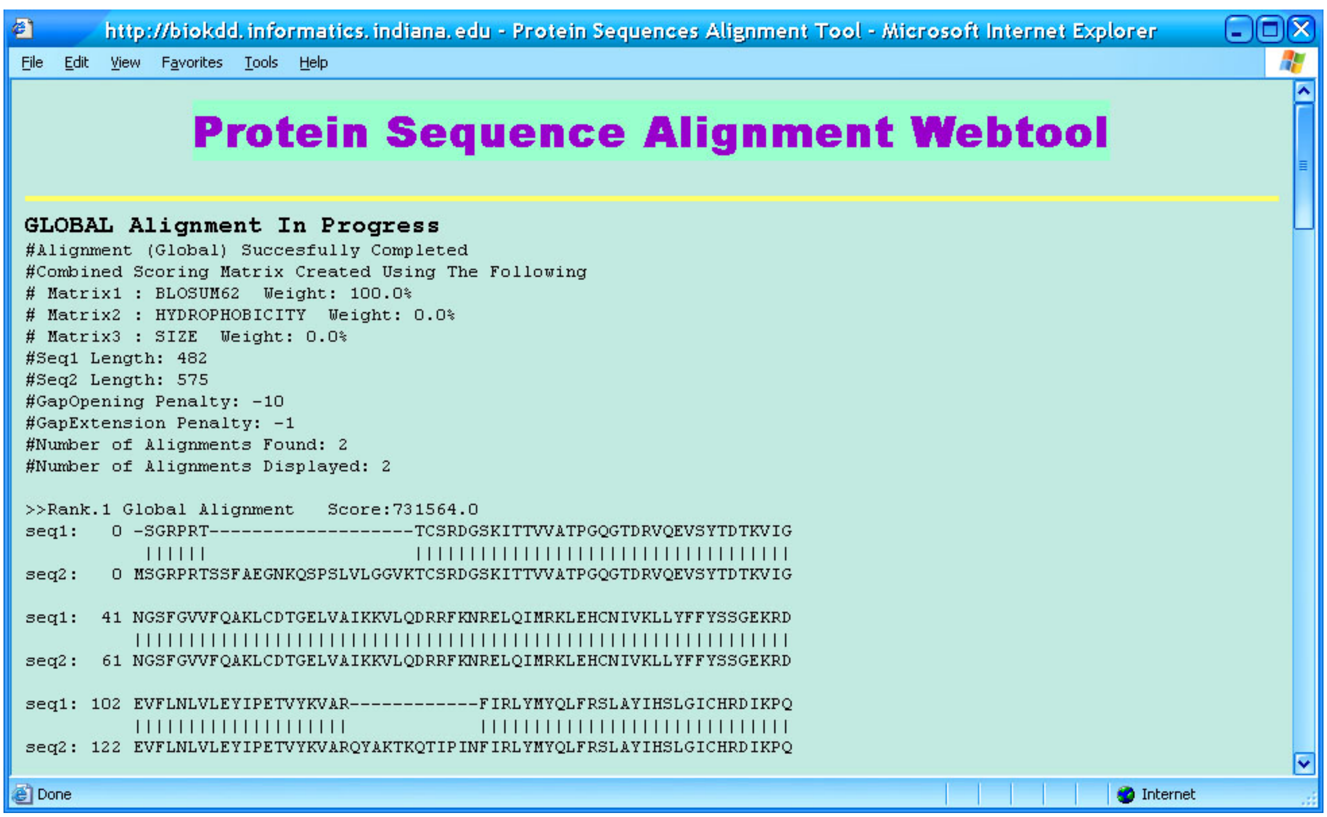
① GLOBAL : Download the global alignment result

② LOCAL : Download the local alignment result

③ BOTH : Download both the global and local alignment results

5. Main page and sample result page





*** Accessible at**

http://biokdd.informatics.indiana.edu/~juhur/L519/HW5/SeqAlign.html

*** Note on the Size matrix.**

The matrix was created by me. It's based upon the physical property of amino acids and the source is http://www.russell.embl-heidelberg.de/aas/aas.html by Robert B. Russell, Matthew J. Betts & Michael R. Barnes

20 Amino acids were grouped into three categories.

1. Tiny : A, G, C, S
2. Small : P, N, D, T, V
3. Large : E, F, H, I, K, L, Q, R, W, Y, M

The scoring scheme is as following

Self	<->	Self	: 10
Tiny	<->	Tiny	: 5
Tiny	<->	Small	: 2
Tiny	<->	Large	: -5
Small	<->	Small	: 5
Small	<->	Large	: -2
Large	<->	Large	: 5

This scheme is based upon my personal insight and a simple assumption that same group can be acceptably substituted but substitutions between tiny(small) and large groups are generally unfavorable.